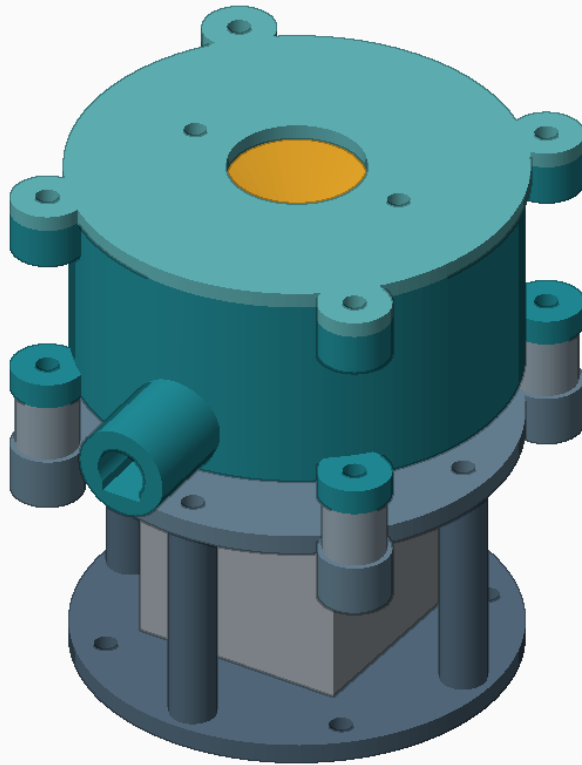


Central Trap / compact v2



HALF-SIZE BOWL

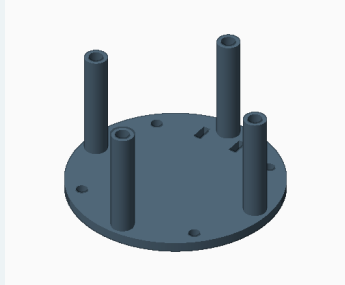
64 mm diameter, reduced from 128 mm. Approximate assembled envelope: 86 x 103 x 104 mm, including the drain.

A small, modular experiment for central collection of fine heavy particles. Full-size M4 hardware and the NEMA17 motor are retained.

Read before printing
Check motor dimensions and insert fit on pages 3-4. This revision has analytical screening and CAD checks, not CFD or demonstrated gold recovery.

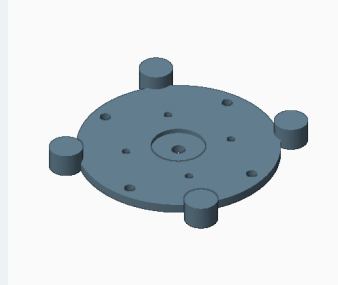
Blue arrows show assembly motion. Gold parts rotate. Teal parts catch water. Rendered screws are replaced by fastener maps and size labels.

Printed parts / one small kit



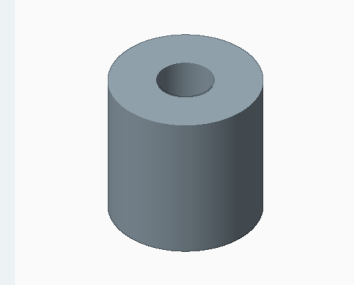
P01 Base

x1



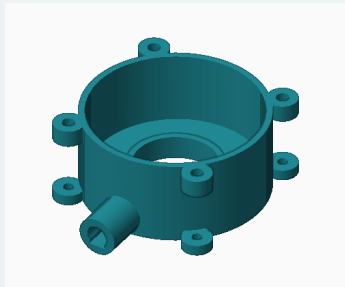
P02 Motor deck

x1



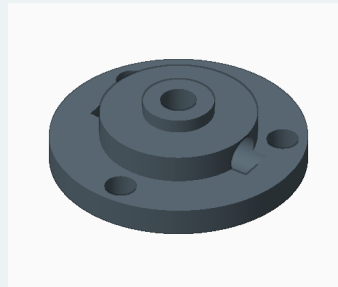
P03 Spacer

x4



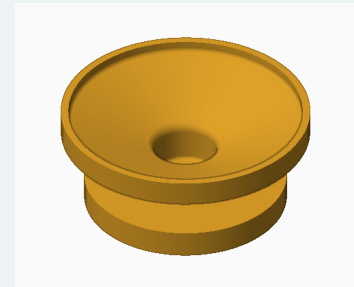
P04 Catcher

x1



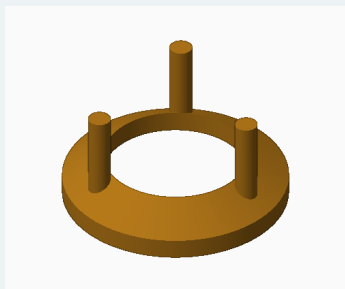
P05 Drive hub

x1



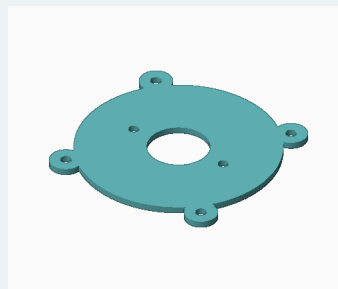
P06 Bowl

x1



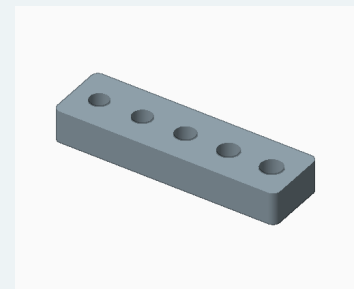
P07 Optional insert

x1



P08 Cover

x1



P09 Fit coupon

x1

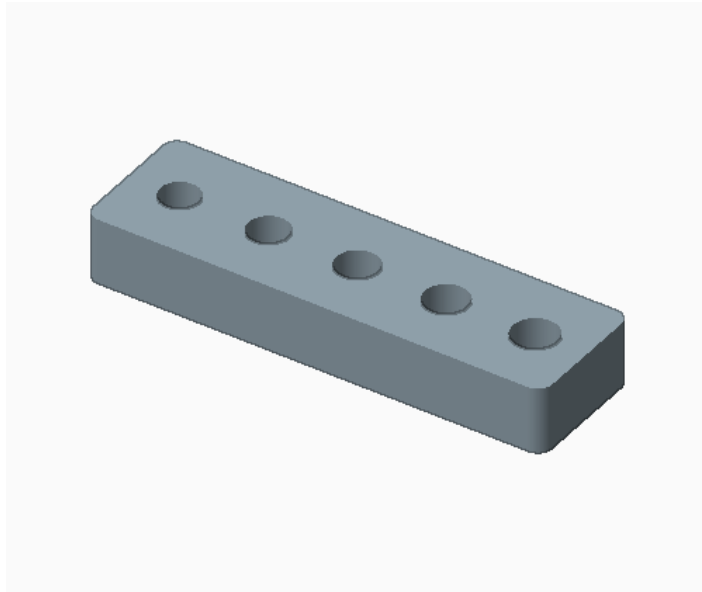
Hardware / keep these sizes full-size

M4 heat-set inserts	17	About 6 OD x 6 long; test coupon first
M4 x 10 socket screws	4	Motor deck to base
M4 x 25 socket screws	4	Catcher through spacers into deck
M4 x 8 socket screws	7	3 hub-to-bowl + 4 cover
M4 x 10 flat-point set screws	2	Opposed motor-shaft locking screws
M3 x 8 socket screws	4	Motor mounting; verify thread depth
F5-10M / F5-10G thrust bearing	1	5 bore x 10 OD x 4 overall
5 mm shaft NEMA17 motor	1	Body/mounting assumptions listed below
Drain hose + feed tubes	-	18 mm ID drain; 4 mm OD feed tubes

Motor fit assumption: 42 mm square body, 40 mm body length, 31 mm mounting-hole square, 22 mm pilot no more than 2 mm high. Shaft projection above the mounting face must be about 18-26 mm. A 5 mm shaft alone does not verify these dimensions.

Also needed: matching hex keys, soldering iron with insert tip, calipers, cable ties, a secondary spill tray, and a current-limited stepper driver/controller. Exact motor current and pinout remain unverified. Do not guess wire pairs or driver current.

Prepare / test the inserts before the full print



5.2 5.4 5.6 5.8 6.0

Coupon holes increase in diameter from the small-hole end. Measure to identify them; this view is illustrative. All assembly insert holes are nominally 5.6 mm.

Press one insert squarely into the coupon. Let it cool, then test a screw. The insert must hold without cracking or spinning. If another hole size fits better, change INS in build_v2.py and regenerate the parts; do not scale the STL.

Before printing: verify motor body, pilot, hole pitch and shaft projection. Unknown motor and insert dimensions are fit assumptions, not measured facts.

INSERT MAP

P01 base

4 / upper ends of posts

P02 deck

4 / outer bosses, from above

P04 catcher

4 / upper cover bosses

P05 hub

2 / opposite radial holes

P06 bowl

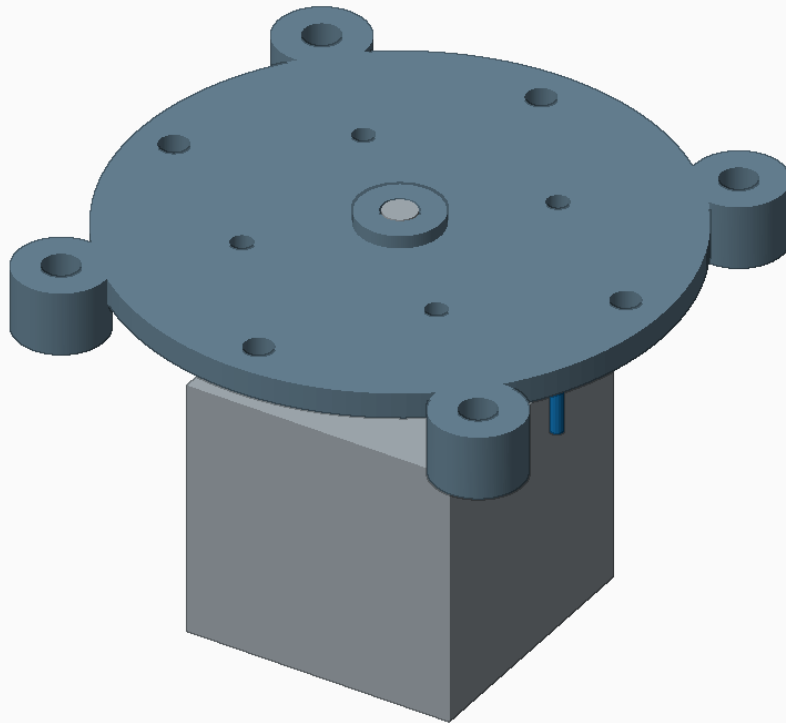
3 / blind underside holes

Seat inserts flush. Keep heat away from bearing seats and the bowl floor. Never drill the bowl holes through into the wet chamber.

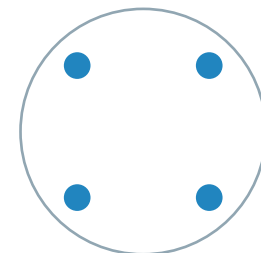
01 Attach the motor to its deck

ADD P02 + motor

4 x M3 x 8 socket screws



- A** Face the motor shaft upward. Put the motor pilot into the recess on the underside of P02.
- B** Install four M3 screws from above the deck. Tighten evenly. Use shorter screws if the motor thread depth is insufficient.

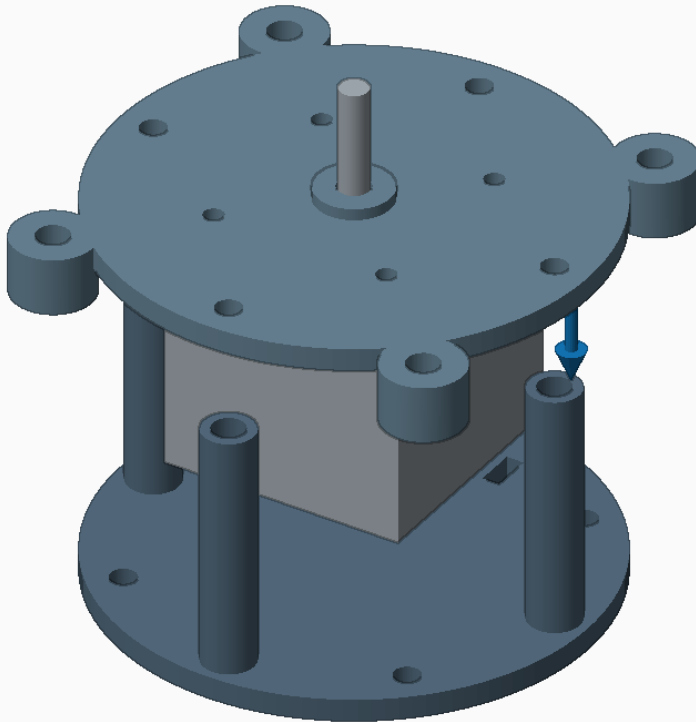
**FASTENER POSITIONS**

CHECK Motor face sits flat. Shaft passes freely through the 5.6 mm opening.

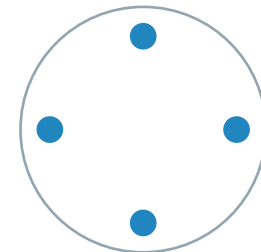
02 Lower the motor into the base

ADD P01 + motor/deck assembly

4 x M4 x 10 socket screws



- A** Lower the motor between the four base posts. Align the four inner deck holes with the post inserts.
- B** Run the cable out through an open side. Tie it to the two base slots, clear of the shaft and drain side.
- C** Install the four M4 screws. Secure the base to the bench through its spare mounting holes.



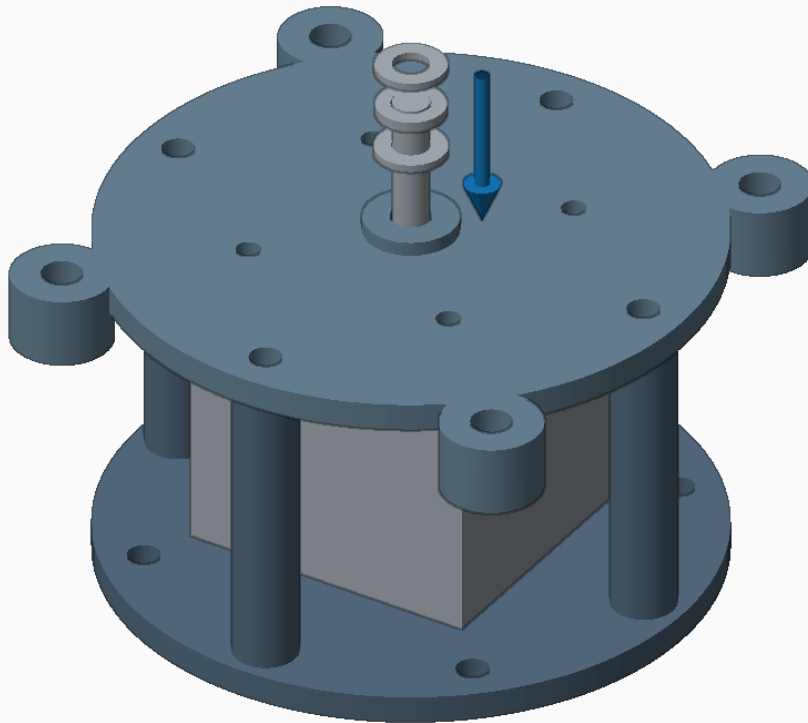
FASTENER POSITIONS

CHECK Motor clears the base floor; wires are not pinched.

03 Stack the thrust bearing

ADD 1 x complete 5 x 10 x 4 thrust bearing

No screws in this step



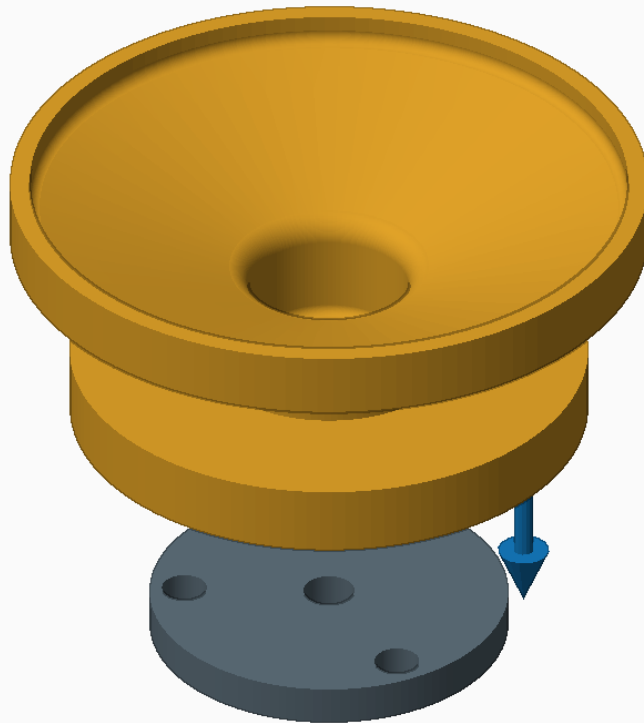
- A** Slide the lower housing washer onto the central pedestal, followed by the ball cage and upper shaft washer.
- B** Bearing grooves face the balls. Use the bearing maker's orientation; the housing washer typically has the slightly larger bore.
- C** The upper washer turns with the hub. This bearing carries axial load; the motor still guides the rotor radially.

CHECK Bearing stack sits flat on the pedestal and turns freely.

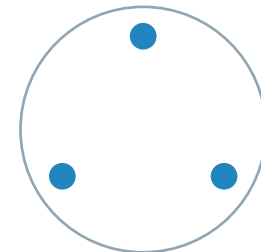
04 Build the rotor outside the machine

ADD P05 hub + P06 bowl

3 x M4 x 8 socket screws; 2 x M4 x 10 set screws



- A** Align the three hub flange holes with the three dry inserts under the bowl.
- B** Insert the three M4 x 8 screws from below the hub flange. Tighten evenly without distorting the bowl.
- C** Start both radial set screws in the hub. Back them out until the shaft bore is unobstructed.



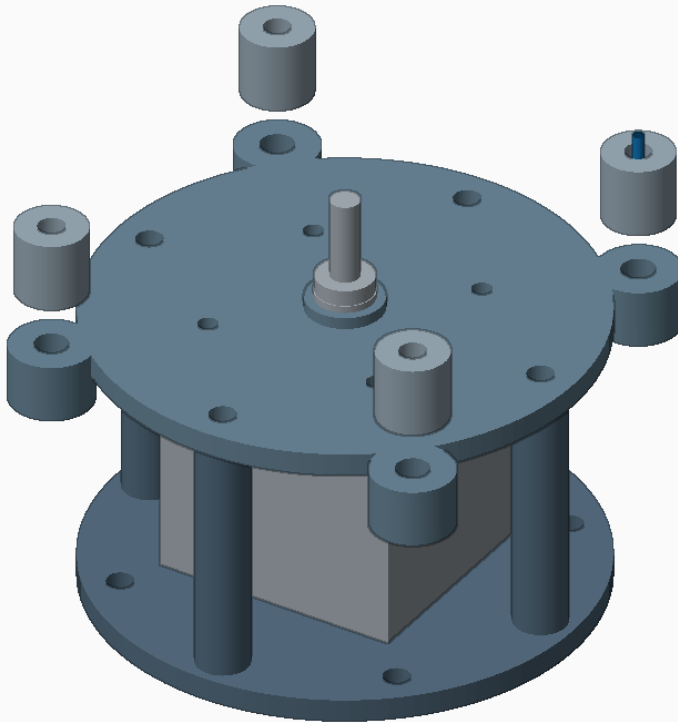
FASTENER POSITIONS

CHECK No screw enters the water chamber. Keep this rotor aside for step 7.

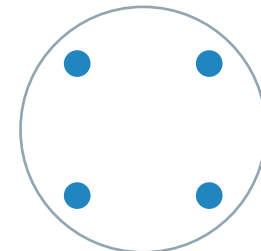
05 Set the four spacers

ADD 4 x P03 spacers

No screws yet



- A** Place one 12 mm spacer on each of the four outer deck bosses.
- B** Align the spacer holes with the inserts below. Keep the bearing and motor shaft clean.



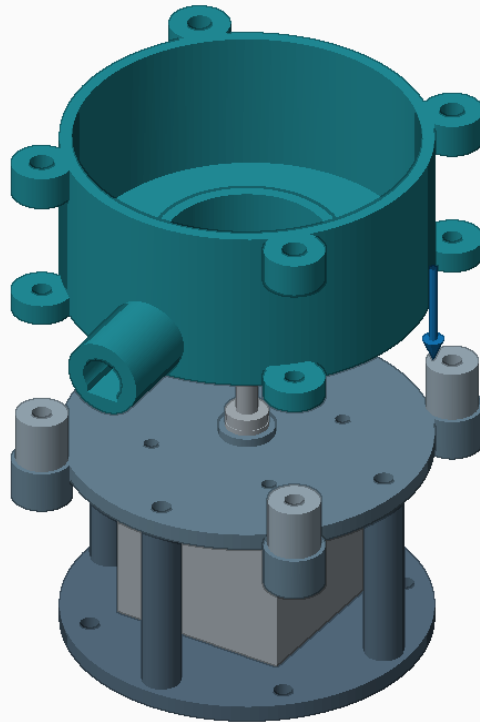
FASTENER POSITIONS

CHECK All four spacers sit upright and flat.

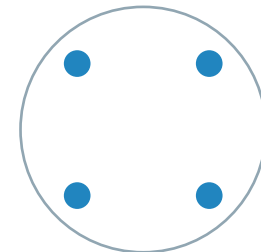
06 Mount the catcher

ADD P04 catcher

4 x M4 x 25 socket screws



- A** Lower the catcher onto the four spacers. Point the drain toward your collection hose.
- B** Pass each screw through a lower catcher tab and spacer into a deck insert. Tighten in alternating pairs.
- C** The large central opening stays open for installation of the rotor. Keep the upper cover inserts empty.

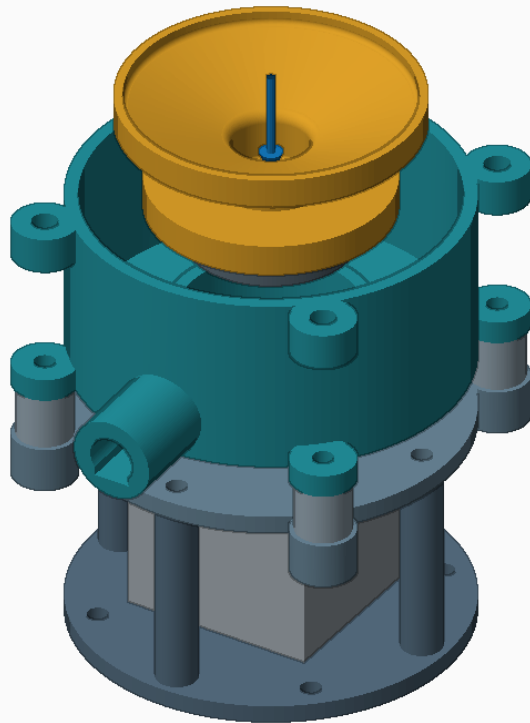
**FASTENER POSITIONS**

CHECK Four spacers are trapped squarely. The catcher is stationary.

07 Seat and lock the rotor

ADD Preassembled P05 + P06 rotor

Tighten the 2 opposed M4 set screws



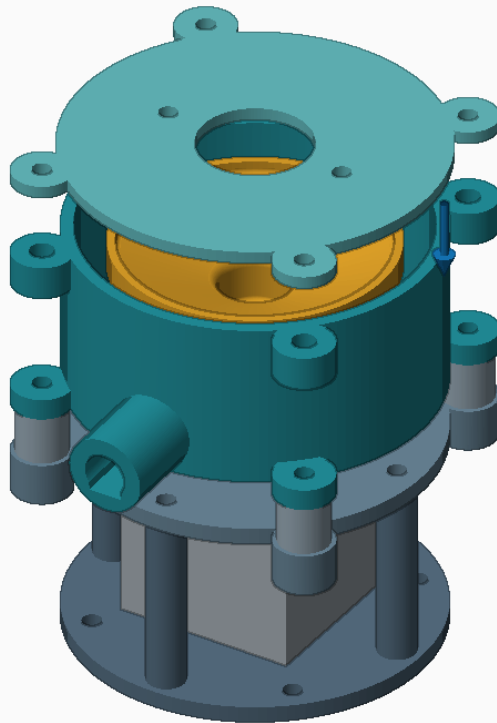
- A** Lower the rotor onto the 5 mm shaft. Its hub nose rests on the upper thrust-bearing washer.
- B** Reach the two set screws through the gap below the catcher. Tighten evenly with a straight hex key; do not force axial preload.
- C** Turn the bowl by hand through a full revolution. Stop if it rubs, rocks or binds. Adjust fit before powering.

CHECK Nominal skirt clearance: 1.3 mm radial, 1 mm vertical. Real print runout must fit inside it.

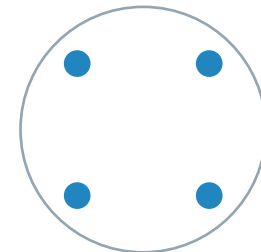
08 Fit the cover and water lines

ADD P08 cover + hose + feed tube

4 x M4 x 8 socket screws



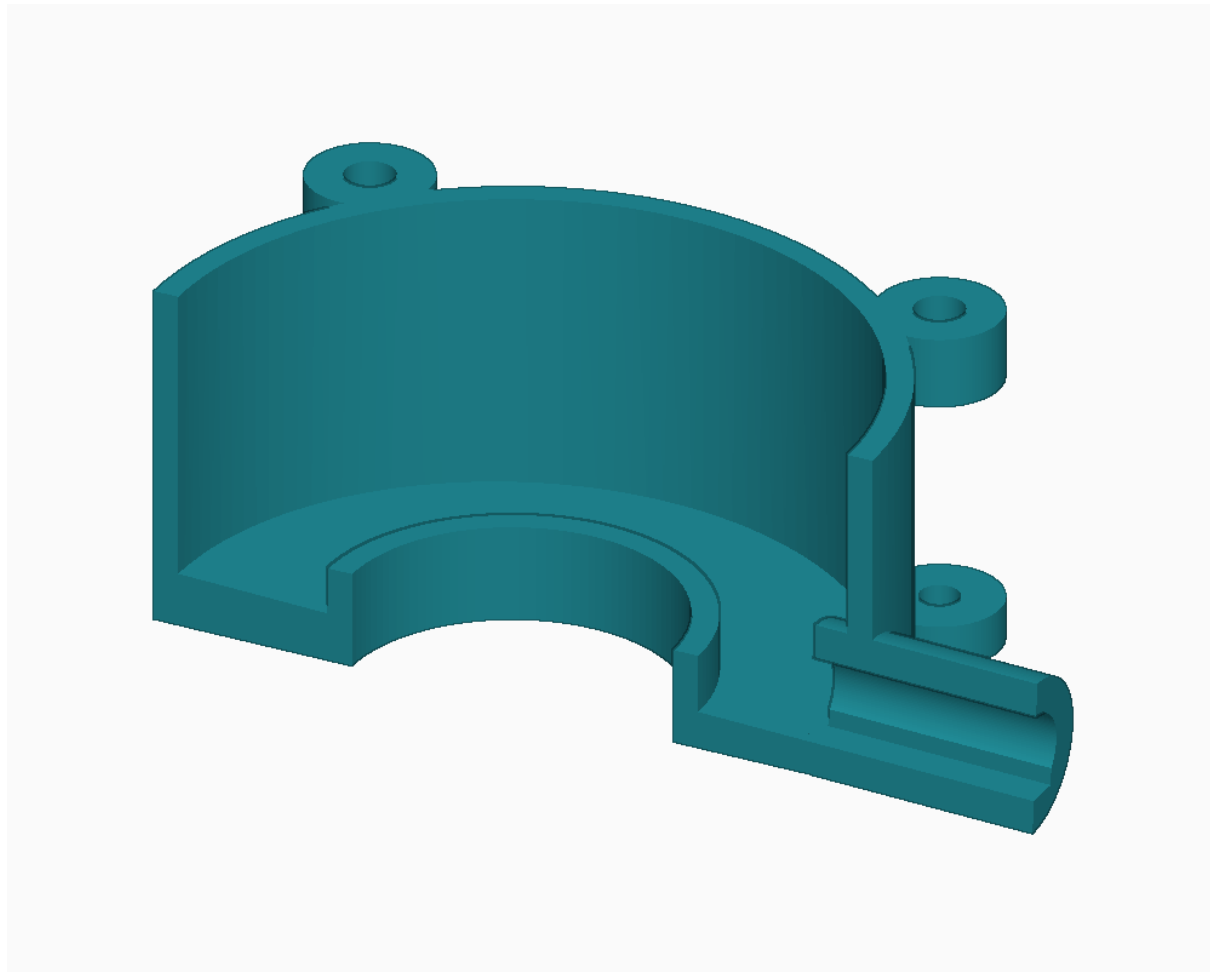
- A** Install the cover on the four upper catcher bosses. Tighten the four screws lightly and evenly.
- B** Secure feed tubing in the 4.3 mm ports. Keep tube ends at least 3 mm above the highest bowl rim; stop them from sliding downward.
- C** Connect a drain hose to the 18 mm OD spout. Support the hose so it cannot twist the printed spout.



FASTENER POSITIONS

CHECK Feed opening is 26 mm. The lid is a splash cover, not a sealed pressure lid.

Water path / floor-level drain



SECTION THROUGH CATCHER

1 / Overflow

Water passes over the bowl rim into the stationary catcher. The rotating bowl itself has a continuous wet wall and floor.

2 / Sloped floor

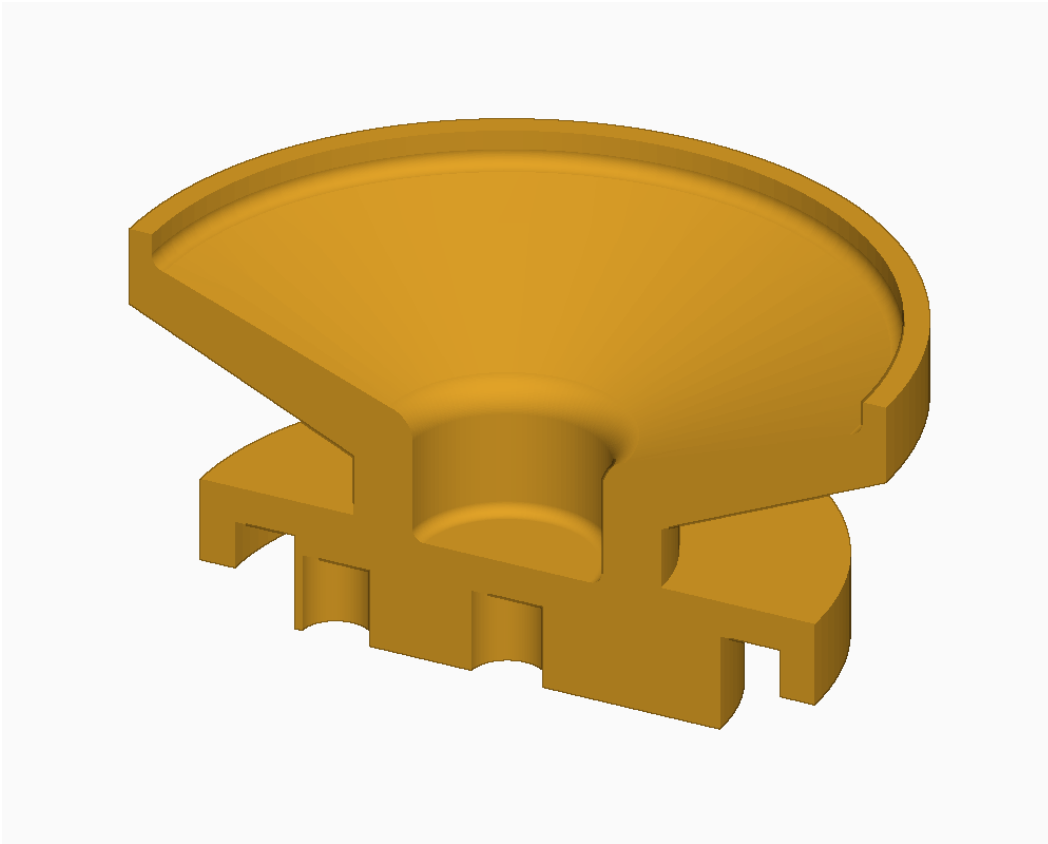
The catcher floor falls 3% toward the drain. The flat-bottomed outlet follows that slope, with no designed raised sill.

3 / Downhill outlet

Run the hose continuously downhill into a second container. A blocked outlet can flood the central shaft opening and motor.

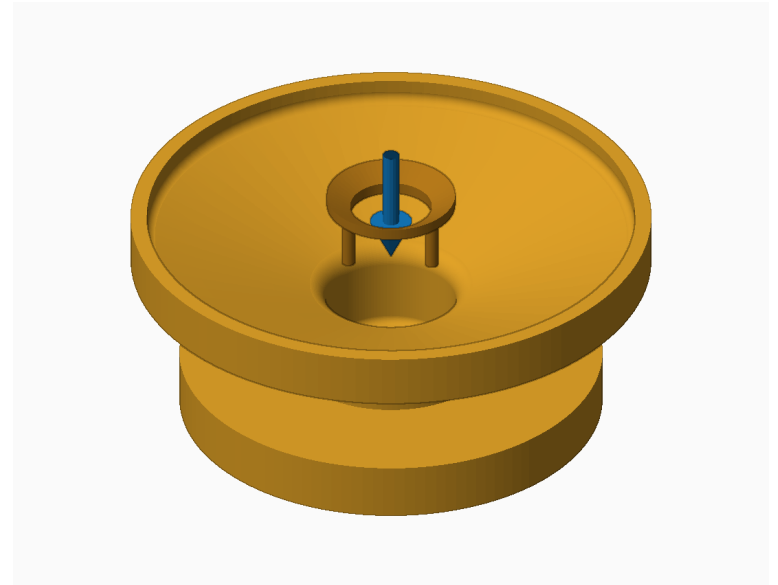
Leak-test bowl and catcher separately before assembly. Closed STL geometry does not make an FDM print watertight. Inspect seams and coat/reprint if needed.

Central sump / start without the optional insert



Open sump - first test

The 16 mm sump has a smooth rounded entrance and roughly 1.7 mL of deep pocket volume. There is no raised collar across the working slope. The slope may help migration only when grains can actually slip or be mobilized.



P07 - later A/B test only

Three feet rest on the sump floor. The cone remains below the entrance; its central opening is 10 mm. It is gravity-seated, not locked. Omit it if warped, loose or lifting during a test.

A pocket can trap sand and clay as well as gold. This is not a one-way valve and cannot guarantee that fine gold stays in. Inspect and clean it frequently.

Printing / use the delivered orientations

STLs are in millimeters and already oriented for printing. Print one of each file, except P03: print four. Do not apply 50% scaling in the slicer; the redesign is already reduced.

P01	Base down	Brim helps tall posts; keep post bores clean
P02	Deck inverted	Support broad deck face around the 2 mm bearing pedestal
P03	Spacer upright	No support normally needed
P04	Catcher upright	Support spout, lower body and projecting upper bosses
P05	Flange down	Keep shaft bore and radial insert bores clear
P06	Bowl upright	Support external flare and underside skirt; protect wet face
P07	Cone inverted	Fine layers; delicate legs; experimental part
P08	Cover flat	No support normally needed
P09	Coupon flat	Print first using the intended final material/settings

Starting settings: PETG, 0.20 mm layers, 5-6 perimeters, 6-8 top/bottom layers, 40-60% infill. Use 0.12-0.16 mm layers for P07. These are starting points, not a waterproofing certification. Remove supports and strings; keep paired rotating hardware equal.

Commission / measure losses before adding more feed

A DRY FIT

Hand-turn a full revolution. Check shaft grip, bearing seating, cover clearance and cable routing. Keep a spill tray underneath.

B WATER ONLY

Start with small oscillations, about ± 3 to 5 degrees at 0.5-1 Hz. Use smooth acceleration. Check leakage and rubbing. These are exploratory commands, not a motor rating.

C SMALL SAMPLE

Wet-screen and disperse clay. Begin with 0.5-2 g of screened sand, not a full hopper. Try 5-30 mL/min water after filling, then adjust to obtain gentle overflow. Stop if material packs or the drain backs up.

D KEEP EVERY TAILING

Use separate containers for overflow and final concentrate. Rinse and inspect both. Test with known-size gold only after mechanical tests; quantify losses by size class.

E CHANGE ONE VARIABLE

Compare open sump to optional insert; record water, frequency, angle, sample mass and run time. Delay any spin/ejection experiments until baseline recovery is measured.

No unattended continuous-feed claim: capacity, clay handling, recovery and safe motor settings are unproven. Never discard test tailings. No controller or automatic feeder is included.

Mechanics / what has and has not been modeled

ANALYTICAL SCREENING

The working slope is approximately 20.2 degrees. Rotation produces outward acceleration $a = \omega^2 \times \text{radius}$. At 30 rpm and a 29 mm radius, this is about 0.029 g. Faster spinning increasingly opposes travel toward the center.

A simplified particle sliding down a rotating cone requires: $\omega^2 \times r / g < (\tan(\beta) - \mu) / (1 + \mu \tan(\beta))$. Here μ is sliding friction. The ideal dry-contact model omits drag, bed interactions and transient water motion. Density cancels: this is not a gold-selection rule.

Calculated inward-sliding limits at 29 mm radius:

$\mu = 0$: 107 rpm; 0.1: 89 rpm;

0.2: 70 rpm; 0.3: 44 rpm.

At $\mu = 0.4$, the model predicts no spontaneous inward sliding even at rest. These are not operating speed limits.

FINE PARTICLES NEED TIME

Stokes settling gives $v = (\rho_{\text{particle}} - \rho_{\text{water}}) g d^2 / (18 \text{ viscosity})$. A spherical 10 micrometer gold grain in clean water needs roughly 5 seconds to fall 5 mm; a 20 micrometer sphere needs about 1.25 seconds.

At five times the viscosity, those times become about 25 and 6.3 seconds. Real clay slurry may be non-Newtonian; gold flakes, attached clay and a moving particle bed are outside this simple model.

Published tea-leaf-paradox studies support possible inward near-bottom transport under some rotating-flow conditions. They also show sensitivity to geometry and confinement. They do not establish this device as a gold concentrator.

No CFD, particle-transport simulation or bench recovery test was performed. Bowl angle and sump geometry are testable choices, not an identified optimum or a guarantee.

Review record / references and revision notes

GEOMETRY REVIEW

All 9 STLs were rendered and visually reviewed from top-oblique, underside-oblique and front views. The bowl and catcher were also sectioned. The package includes these review sheets.

Every exported STL passed closed-edge, consistent-orientation and positive-volume checks. All CAD parts are valid single solids. The modeled assembly has no detected interference; the rotor was checked every 15 degrees. This does not measure print tolerance, physical balance or recovery.

Changed from v1: 64 mm bowl; shorter direct-shaft drive with a thrust bearing; round mounting bosses; sloped catcher floor and floor-level D-shaped drain; smooth open sump entry; optional recessed insert. The 18 mm hose connection and M4 hardware remain full-size.

PRIMARY REFERENCES

1. Rotating-flow / bottom inclination study, Micromachines 14(11), article 2024 (2023).

[Open primary source](#)

2. Stepper-driven cup / particle aggregation study, volume 4(3), article 37 (2024).

[Open primary source](#)

3. SMB Bearings: F5-10G dimension drawing (5 x 10 x 4 mm).

[Open primary source](#)

4. US EPA: gravity concentration, feed preparation and recovery context.

[Open primary source](#)

Sources explain mechanisms and hardware dimensions. None validates this prototype. Exact Moons C17HD40-102-0 1N ratings were not verified.

Files: STL print parts; STEP geometry; assembly.step; build_v2.py; check reports; multi-view review sheets. Re-run the generator after changing motor or insert dimensions.